

Kubernetes Deployment Challenge: BookStack App

Context:

You're deploying **BookStack**, a documentation platform, along with a **MariaDB** backend, on a Kubernetes cluster. The application should be production-ready, using persistent storage, secure credentials, and accessible via a browser.

Tasks

1. Deploy BookStack and MariaDB in a dedicated namespace.
 2. Ensure sensitive credentials such as database passwords and mail settings are securely handled.
 3. Use persistent storage for both the BookStack configuration and the MariaDB database.
 4. BookStack must connect to the MariaDB instance for database operations.
 5. Mail notifications must be configured using SMTP (Gmail).
 6. The BookStack web interface must be exposed and reachable externally.
 7. Verify the deployment works end-to-end:
 - BookStack loads in a browser.
 - The database connection is successful.
 - Emails can be sent from the app.
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Success Criteria

- Pods are running and healthy.
- Persistent volumes are mounted and in use.
- Secrets are not exposed in plain text.
- Services allow internal and external connectivity.
- BookStack loads correctly and functions fully.

Kubernetes Deployment Reference:

BookStack Stack

This document outlines the required components and configurations needed to deploy the BookStack application and its MariaDB database backend to a Kubernetes cluster. It is based on an analysis of a Docker Compose setup and serves as a conversion reference for Kubernetes deployments.

Main Configuration Overview

Component	Type	Details
App Name	Application	BookStack (lscr.io/linuxserver/bookstack:latest)
Database	Service	MariaDB (lscr.io/linuxserver/mariadb)
App Port	Port	80 (exposed as 6875:80 in Compose)
Volumes	Persistent Storage	bookstack_config:/config, bookstack_db_data:/var/lib/mysql
Network	Internal Network	proxy_network (or default)
Environment Vars	Configuration	See environment variables section below

Environment Variables - BookStack

Variable	Value
PUID	1000
PGID	1000
TZ	Etc/UTC
APP_URL	https://bookstack.edusuc.net
APP_KEY	base64:6um2Xi48STZ+dqwdHJOCgta1/VKJhdUj4gFjZEfchu4=
DB_HOST	bookstack_db
DB_PORT	3306
DB_USERNAME	bookstack
DB_PASSWORD	bookstack

DB_DATABASE	bookstackapp
MAIL_DRIVER	smtp
MAIL_FROM_NAME	BookStack
MAIL_FROM	info@devopseasylearning.com
MAIL_HOST	smtp.gmail.com
MAIL_PORT	465
MAIL_USERNAME	info@devopseasylearning.com
MAIL_PASSWORD	Edhfgnurn\$@!HDK
MAIL_ENCRYPTION	tls

Environment Variables - MariaDB

Variable	Value
PUID	1000
PGID	1000
TZ	Etc/UTC
MYSQL_ROOT_PASSWORD	bookstack
MYSQL_DATABASE	bookstackapp
MYSQL_USER	bookstack
MYSQL_PASSWORD	bookstack